ANALYSIS RECORD

Sankey Diagram: Travel-Booking Website Sessions

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Date	13 October 2026

Scenario

The analytics team of a travel-booking website tracked 3,900 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".

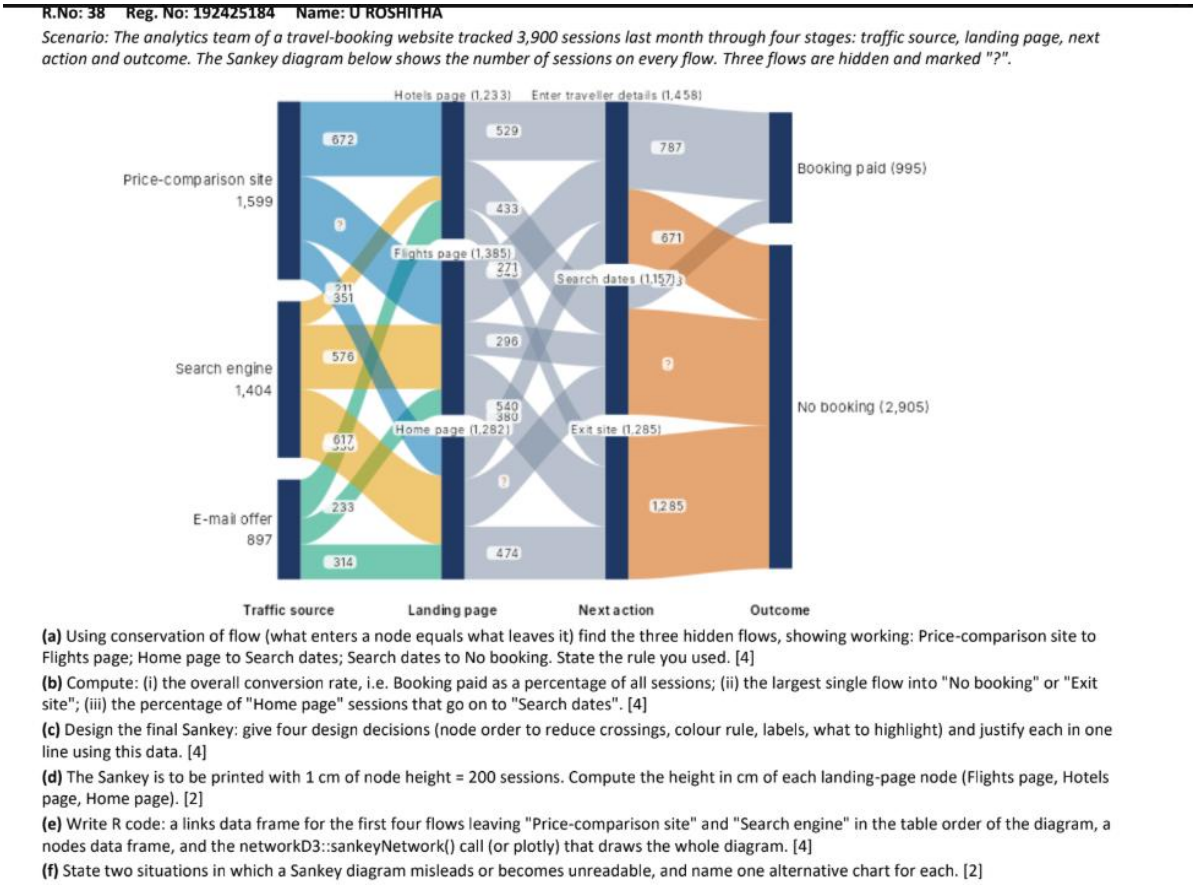


Figure 1: Sankey diagram of sessions (given)

Data read from the diagram

Each of the four stages contains all 3,900 sessions, so the node totals in every column add up to 3,900.

Stage	Nodes and totals	Sum
Traffic source	Price-comparison site 1,599; Search engine 1,404; E-mail offer 897	3,900
Landing page	Hotels page 1,233; Flights page 1,385; Home page 1,282	3,900
Next action	Enter traveller details 1,458; Search dates 1,157; Exit site 1,285	3,900
Outcome	Booking paid 995; No booking 2,905	3,900

(a) Hidden flows by conservation of flow [4]

Rule used: conservation of flow. At every intermediate node, the sessions that enter the node equal the sessions that leave it, and both equal the node total. Therefore a missing flow = node total minus the sum of the other flows on the same side of that node.

Flow 1: Price-comparison site to Flights page

Price-comparison site has 1,599 sessions leaving it. The labelled flows are 672 (to Hotels page) and 351 (to Home page).

Price-comparison site to Flights page = 1,599 - 672 - 351 = 576.

Check using the Flights page inflow: 1,385 - 576 (Search engine) - 233 (E-mail offer) = 576. Both methods agree.

Flow 2: Home page to Search dates

Home page has 1,282 sessions leaving it. The labelled flows are 380 (to Enter traveller details) and 474 (to Exit site).

Home page to Search dates = 1,282 - 380 - 474 = 428.

Check using the Search dates inflow: 1,157 - 433 (Hotels page) - 296 (Flights page) = 428. Both methods agree.

Flow 3: Search dates to No booking

Booking paid receives 995 sessions. The labelled flow from Enter traveller details is 787, so Search dates to Booking paid = 995 - 787 = 208 (the thin unlabelled flow).

Search dates to No booking = 1,157 - 208 = 949.

Check using the No booking inflow: 2,905 - 671 (Enter traveller details) - 1,285 (Exit site) = 949. Both methods agree.

Complete flow table (hidden flows in bold)

Flow	Sessions	Flow	Sessions
Price-comparison to Hotels	672	Flights to Search dates	296
Price-comparison to Flights	**576**	Flights to Exit site	540
Price-comparison to Home	351	Home to Enter details	380
Search engine to Hotels	211	**Home to Search dates**	**428**
Search engine to Flights	576	Home to Exit site	474
Search engine to Home	617	Enter details to Booking paid	787
E-mail to Hotels	350	Enter details to No booking	671

Flow	Sessions	Flow	Sessions
E-mail to Flights	233	Search dates to Booking paid	208
E-mail to Home	314	**Search dates to No booking**	**949**
Hotels to Enter details	529	Exit site to No booking	1,285
Hotels to Search dates	433	Flights to Enter details	549
Hotels to Exit site	271	Total of each stage	3,900

Note: the label beneath 271 is overprinted in the image. The Flights page to Enter traveller details flow (549) follows from the same rule: 1,458 - 529 - 380 = 549, and 549 + 296 + 540 = 1,385.

(b) Calculations [4]

(i) Overall conversion rate

Conversion rate = Booking paid / all sessions x 100 = 995 / 3,900 x 100 = 25.51% (about 25.5%).

(ii) Largest single flow into "No booking" or "Exit site"

Flows into Exit site: 271, 540 and 474. Flows into No booking: 671, 949 and 1,285. The largest is Exit site to No booking = 1,285 sessions, which is 1,285 / 3,900 = 32.95% (about 33%) of all sessions.

(iii) Home page sessions that go on to Search dates

428 / 1,282 x 100 = 33.4% (about one in three Home page sessions).

(c) Final Sankey design decisions [4]

Decision	Choice	Justification (using this data)
Node order	Order each column top to bottom as Booking paid, Enter details / Search dates / Exit site, and sort landing pages by their largest next action.	Enter details (787 booking) sits at the top and Exit site (1,285 no booking) at the bottom, so those two big flows run almost straight with fewer crossings.
Colour rule	Colour links by traffic source (blue, yellow, green) and use grey for later flows; keep nodes in neutral grey.	Only three sources, so three colours stay readable and show where each landing-page flow came from.
Labels	Label each node with its name, sessions and share of 3,900 (for example, Exit site 1,285 / 33%); label links of 200 sessions or more outside the bands.	The smallest flow is 208 and the largest is 1,285, so every label fits without the overprinting seen on 271 and 549.
Highlight	Highlight Enter details to Booking paid (787) and the Exit site to No booking leak (1,285) in strong colours and fade the rest.	787 of 995 bookings (79.1%) come through Enter details, while 1,285 sessions (33%) leave without any action.

(d) Node heights when 1 cm = 200 sessions [4]

Height (cm) = sessions / 200.

Landing-page node	Sessions	Working	Height (cm)
Flights page	1,385	1,385 / 200	6.925 cm
Hotels page	1,233	1,233 / 200	6.165 cm
Home page	1,282	1,282 / 200	6.410 cm

Check: 6.925 + 6.165 + 6.410 = 19.5 cm = 3,900 / 200, so the three nodes together match the full column height (excluding gaps between nodes).

(e) R code [4]

The first four rows of the links data frame are the first four flows leaving "Price-comparison site" and "Search engine" in the table order of the diagram (top to bottom): Price-comparison to Hotels, Flights and Home, then Search engine to Hotels. The remaining rows are added so that the whole diagram is drawn. Node numbers start at 0, as networkD3 requires.

```
library(networkD3)

# Nodes: row position (starting at 0) is the node number
nodes <- data.frame(name = c(
  "Price-comparison site", "Search engine", "E-mail offer",   # 0-2 Traffic source
  "Hotels page", "Flights page", "Home page",                 # 3-5 Landing page
  "Enter traveller details", "Search dates", "Exit site",     # 6-8 Next action
  "Booking paid", "No booking"))                              # 9-10 Outcome

# Links: source, target, value (sessions)
links <- data.frame(
  source = c(0, 0, 0,  1, 1, 1,  2, 2, 2,  3, 3, 3,
             4, 4, 4,  5, 5, 5,  6, 6,  7, 7,  8),
  target = c(3, 4, 5,  3, 4, 5,  3, 4, 5,  6, 7, 8,
             6, 7, 8,  6, 7, 8,  9, 10,  9, 10, 10),
  value  = c(672, 576, 351,  211, 576, 617,  350, 233, 314,
             529, 433, 271,  549, 296, 540,  380, 428, 474,
             787, 671,  208, 949,  1285),
  group  = c(rep("Price", 3), rep("Search", 3), rep("Email", 3),
             rep("Other", 14)))

# Sanity check: every stage must add up to 3,900 sessions
stopifnot(sum(links$value[1:9]) == 3900,
          sum(links$value[10:18]) == 3900,
          sum(links$value[19:23]) == 3900)

# Draw the whole diagram
sankeyNetwork(Links = links, Nodes = nodes,
              Source = "source", Target = "target", Value = "value",
              NodeID = "name", LinkGroup = "group",
              colourScale = JS('d3.scaleOrdinal()
                .domain(["Price","Search","Email","Other"])
                .range(["#5DA5DA","#F2C14E","#5CBFA9","#C8C8C8"])'),
              units = "sessions", fontSize = 13,
```

```
        fontFamily = "Times New Roman",
        nodeWidth = 25, nodePadding = 20,
        iterations = 0, sinksRight = FALSE)
```

Note: iterations = 0 keeps the nodes in the order of the nodes table instead of letting the layout reorder them.

(f) When a Sankey diagram misleads or becomes unreadable [2]

Situation	Why it fails	Alternative chart
Too many nodes and flows (for example, dozens of landing pages and sources)	Bands cross heavily, labels overlap and the eye cannot follow any single flow.	Heatmap (matrix) of source against landing page
The question is about rates or small flows (for example, conversion % or the 208-session flow)	Band width shows only absolute counts, so conversion rates are hard to compare and thin flows almost vanish.	Funnel chart or 100% stacked bar chart

Summary of answers

Part	Answer
(a)	Price-comparison to Flights = 576; Home to Search dates = 428; Search dates to No booking = 949. Rule: inflow = outflow at every node.
(b)	(i) 25.51% (ii) Exit site to No booking, 1,285 sessions (iii) 33.4%
(c)	Node order by dominant flow; colour by source; labelled nodes with share of sessions; highlight 787 booking path and 1,285 exit leak.
(d)	Flights 6.925 cm; Hotels 6.165 cm; Home 6.410 cm.
(e)	networkD3 code with nodes and 23-row links data frame given above.
(f)	Too many nodes (use heatmap); rates and small flows (use funnel or 100% stacked bar).